SOFTWARE ENGINEERING LAB 05

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Static Code Analysis — Final Results

Project: Scholarship Screener

Repository: https://github.com/SETAPESU26/L5_11_scholarship_screener

File analyzed: scholarship_system.py

1. Pylint

Command executed:

```
python -m pylint scholarship_system.py
```

Output:

```
Your code has been rated at 10.00/10
(previous run: 9.91/10, +0.09)
```

Result: The final code achieved a Pylint score of 10.00/10.

2. Flake8

Command executed:

```
python -m flake8 scholarship_system.py
```

Output:

```
(No output)
```

Result: Flake8 reported no style or formatting errors. The empty output file is expected when no errors are reported.

3. Bandit

Command executed:

```
python -m bandit scholarship_system.py
```

Output:

```
Test results:
    No issues identified.

Code scanned:
    Total lines of code: 168
    Total lines skipped (#nosec): 0

Run metrics:
    Total issues (by severity):
        Undefined: 0
        Low: 0
        Medium: 0
        High: 0

    Total issues (by confidence):
        Undefined: 0
        Low: 0
        Medium: 0
        High: 0

Files skipped (0):
```

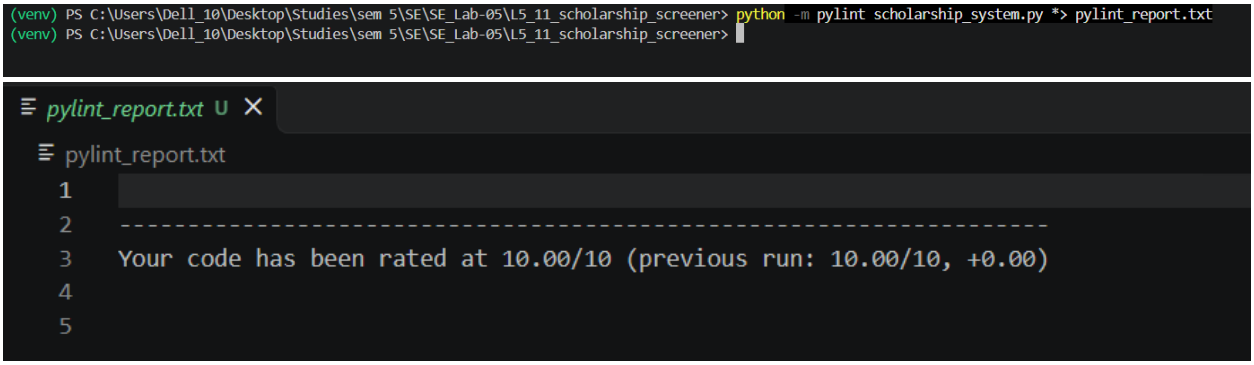
Result: Bandit did not identify any security issues in the scanned file.

Conclusion

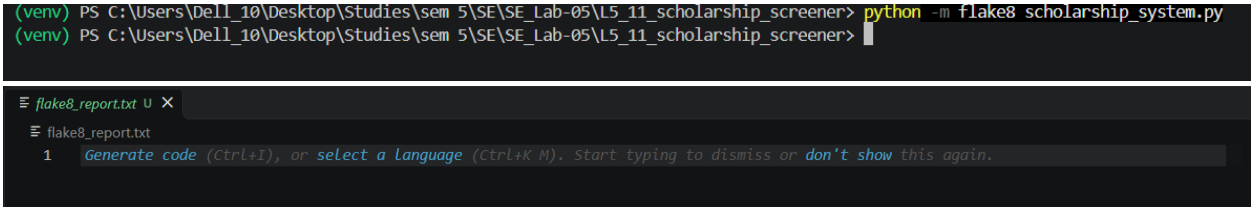
The final version of `scholarship_system.py` achieved a Pylint score of 10.00/10, passed Flake8 without reported errors, and had no issues identified by Bandit. These tools helped assess the code's quality, style, and potential security risks.

Terminal Evidence (Screenshots)

1. Pylint



2. Flake8



No output, since everything is working fine.

3. Bandit

```
(venv) PS C:\Users\Dell_10\Desktop\Studies\sem 5\SE\SE_Lab-05\L5_11_scholarship_screener> python -m bandit scholarship_system.py
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.14.3
Run started:2026-10-20 16:53:50.645908+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 168
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
```

bandit_report.txt U

bandit_report.txt

```
1   Run started:2026-10-20 16:30:06.407464+00:00
2
3   Test results:
4       No issues identified.
5
6   Code scanned:
7       Total lines of code: 168
8       Total lines skipped (#nosec): 0
9       Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0
10
11  Run metrics:
12      Total issues (by severity):
13          Undefined: 0
14          Low: 0
15          Medium: 0
16          High: 0
17      Total issues (by confidence):
18          Undefined: 0
19          Low: 0
20          Medium: 0
21          High: 0
22  Files skipped (0):
23
```